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Dual-Porosity Modeling of Multicomponent Diffusion in Tight Fractured Systems

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Abstract

In recent years, modeling studies of carbon dioxide (CO₂) injection in tight fractured reservoirs have employed an advective/diffusion model coupled with either a single-porosity model (SPM) or an embedded discrete fracture model (EDFM). A key feature of these gridding approaches is their reliance on a large number of gridblocks to represent fracture segments and capture the transient behavior observed in such reservoirs. Incorporating diffusive fluxes into conventional advective-only reservoir simulators adds significant computational complexity, making these approaches particularly demanding.

In this paper, we address the application of the dual-porosity model (DPM) to study diffusive mass transfer in tight fractured reservoirs. The implicit and coarse representation of the fractures in a DPM offers higher computational efficiency compared to the SPM or EDFM. To address the inaccuracies that may result from conventional transfer functions of DPM's, we propose an improved matrix/fracture representation that accounts for both advective and diffusive mass transfer. Both mechanisms follow the functional form of the [Warren and Root \(1963\)](#) model and incorporate a time-dependent correction factor derived from the generalized Vermeulen transfer function ([Zhang et al., 2022](#)). However, the representation of molecular diffusion in the DPM includes n_c independent correction factors, as opposed to a scalar quantity for advective transport. Moreover, to identify relevant matrix/fracture interaction parameters (σ and V), we show that the values for diffusion are equivalent to those for the viscous flow expression.

We first demonstrate, using numerical examples, the validity of the DP compositional model by comparing calculation results with an analytical solution and fine-grid SPM results. These examples serve as a benchmark for the new model in diffusion-dominated problems involving binary and multicomponent fluid systems. Subsequently, we present our efforts to model CO₂ HnP conducted on an Eagle Ford shale core. In all calculations, the DPM approach employs a single-block representation, which outperforms other modeling approaches (SPM and EDFM) that require a larger number of gridblocks.

We demonstrate that the proposed DPM, incorporating an advanced transfer function, provides for an accurate and efficient representation of multicomponent diffusive mass transfer in tight fractured systems. Consequently, this approach serves as a practical and reliable platform for modeling EOR in large-scale

applications by overcoming the limitations of conventional DPM's and avoiding the need for fine gridding required in SPM or EDFM.

Introduction

Tight fractured reservoirs hold significant potential for hydrocarbon production, offering an opportunity to meet growing energy demands. Despite ongoing efforts to develop these resources, a substantial portion of the oil remains unrecovered. Enhanced oil recovery (EOR) presents a promising method to unlock the untapped potential of unconventional reservoirs, with investigators demonstrating attractive incremental recovery from CO₂ injection into tight fractured cores (Darvish et al., 2006; Alharthy et al., 2015; Hoffman, 2018; Alahmari et al., 2023). This strategy has also been evaluated in several field pilot tests, yielding positive results (Hoffman, 2018).

The success of CO₂ injection can be attributed, in part, to molecular diffusion, which may play a critical role in the recovery of oil from tight matrix formations. Therefore, accurate modeling of molecular diffusion is essential to support ongoing field activities and to encourage further investments. To address this, numerous efforts reported in the literature have focused on this topic. However, these efforts often rely on computationally demanding models that are impractical for large-scale studies or lack proper representation of matrix/fracture mass transfer formulation. Below, we provide a brief overview of the mainstream methods for modeling gas injection into fractured systems.

Single-Porosity Models

Traditionally, a SPM is commonly used to conduct core-scale and single-well studies of EOR operations in tight fractured systems (e.g., Vega et al., 2010; Ghasemi et al., 2018; Alahmari et al., 2023). It uses an explicit representation of fractures, where each fracture segment is defined using a set of control volumes. The approach may require local grid refinement at the matrix/fracture interface to capture the transient behavior in the tight matrix. Its advantage of producing accurate results without the need for a matrix/fracture transfer function explains its wide use in many studies. Fine-grid models are considered to provide reference solutions as they accurately represent realistic settings. That being said, a fully resolved solution comes at the cost of computational efficiency, which promotes the use of coarser grids.

Dual-Porosity Models

The DPM can be employed to study tight fractured systems in the context of EOR modeling. This approach is recognized for its efficient simulation capabilities, primarily due to its implicit and coarse representation of the fractures. However, the accuracy of the results is significantly influenced by the selection of the transfer function and the associated shape factor(s).

Shojaei and Jessen (2014) employed a DPM in field-scale studies to illustrate the importance of molecular diffusion during gas injection. Their representation of matrix/fracture mass transfer combined the contributions of fluid expansion, molecular diffusion, and gravity drainage. Each mechanism incorporated a separate time-dependent boost factor, however limited to specific geometries as per findings of Zimmerman et al. (1993) and Lim & Aziz (1995).

Wan and Sheng (2015) employed a Dual-Porosity/Dual-Permeability Model (DPDPM) to investigate the effect of diffusion on oil recovery during gas flooding in two hydraulically fractured horizontal wells, simulating an Eagle Ford shale EOR operation. They attributed the total mass transfer to fluid expansion and molecular diffusion, utilizing the traditional transfer function of Warren and Root (1963), without a boost factor, and the shape factor from Gilman and Kazemi (1983).

Sherafati and Jessen (2018) highlighted the importance of proper representation of diffusive flux in the DPM approach. Their transfer function introduced a correction factor to capture the transient behavior,

noting that a common correction term for all components in a multicomponent mixture would not align well with reference solutions. To address this, they expanded the expression to a vector form, where each element represented the correction term for a specific component based on its concentration. They validated their formulation with experimental data and demonstrated its impact at a larger scale by reporting the results of gas injection into a fractured reservoir.

Jia et al. (2018) examined the role of molecular diffusion during CO₂ HnP in a stimulated horizontal well, emphasizing the impact of reservoir heterogeneity on recovery. They utilized a DPDPM in a commercial simulator (CMG-GEM) for their simulation calculations.

Khaz'ali et al. (2020) developed a fully implicit DP compositional simulator for fractured reservoirs, focusing on modeling of advective and diffusive mass transfer. They introduced a correction factor, embedded in the shape factor calculation, for accurate mass transfer:

$$\text{Advection: } \sigma_A \bar{k} = 4\kappa_A \left(\frac{k}{L^2} \right), \quad (1)$$

$$\text{Diffusion: } \sigma_D = 4\kappa_D \left(\frac{1}{L^2} \right) \quad (2)$$

where σ is the shape factor, k is the permeability, κ is the correction factor, and L is the fracture spacing. They assumed that diffusion is analogous to advection and that the correction factor, for each process, is a dimensionless value to be obtained by history matching. It was not clear if the value is constant or time-dependent.

Fu et al. (2021) conducted experimental and modeling studies to understand the impact of injecting different gases into Eagle Ford cores. Using a DPDPM for numerical modeling, they successfully matched laboratory observations by tuning parameters such as relative permeability, fracture porosity, fracture permeability, and tortuosity.

Sistan et al. (2022) investigated the impact of water presence during CO₂ injection into a fractured core. Their mass transfer representation in the DPM adopted a similar approach to that presented by Shojaei and Jessen (2014).

Embedded Discrete Fracture Models

The EDFM has gained popularity in recent years for modeling EOR in unconventional reservoirs, particularly at the well or larger scales, due to advancements in computational power.

Zuloaga-Molero et al. (2014) utilized the EDFM to study CO₂-EOR, investigating the influence of various fracture geometries and lengths on oil recovery. Their work also examined the impact of reservoir permeability on EOR performance. In modeling molecular diffusion, they assumed a constant CO₂ diffusion coefficient in the oil phase, a simplification that may have limitations in representing complex diffusion dynamics.

Yu et al. (2019) applied the EDFM to simulate HnP operations for an Eagle Ford oil well, which featured 1,800 natural fractures and 19 stages of hydraulic fractures. They calibrated their model by adjusting reservoir parameters to match field production data. Subsequently, they explored nanopore confinement effects and conducted a sensitivity analysis to assess the impact of diffusive mass transfer on recovery. Their assumption that CO₂ diffusion occurs solely in hydrocarbon phases, with a constant diffusion coefficient, represents a limiting simplification in their modeling approach.

A notable drawback of the EDFM is its high computational demands, which make scaling from lab to field-scale models challenging and costly, thereby restricting its widespread application in industry settings. Addressing this limitation may require advancements in computational efficiency or algorithmic improvements to enhance scalability.

In this study, we leverage the efficiency of the classical DPM to enhance the accuracy of diffusive mass transfer modeling in tight fractured reservoirs. Our work distinguishes itself from the aforementioned references by further improving the representation of the mass transfer, distinctly for each recovery mechanism, and expanding the application to more complex problems. Key considerations are: 1) use of an open-source code (MRST) to develop our formulations, contributing to the literature's accessibility; 2) the compositional simulator employs a fully implicit approach; and 3) application of a physics-driven approach to interpret mass transfer in a wide range of geometries and scales.

The manuscript is organized as follows: First, we present the mathematical framework of the new compositional DPM. Subsequently, we validate the model through several experiments, starting with simple fracture geometries where we compare our results against analytical solutions and fine-grid SPM reference solutions. We also propose a practical method for determining the relevant transfer function parameters for irregular fracture geometries. Finally, we present our efforts to model CO₂ HnP conducted on an Eagle Ford shale core, demonstrating its capability to address complex field-scale scenarios.

Mathematical Model

In this study, we employ the MATLAB Reservoir Simulation Toolbox (MRST) to investigate transport mechanisms in tight fractured reservoirs. Building on our earlier work (Raslan et al., 2023b), we develop a comprehensive advection/diffusion model using a DP compositional formulation. This advancement extends the advection-only DP compositional model by incorporating molecular diffusion, enabling a more accurate representation of matrix/fracture mass transfer that accounts for the combined contributions of advective and diffusive transport.

In a DP representation, two coupled differential equations, per component, are used to describe the mass transport in the porous material. We consider the multicomponent mass-balance equations for the fracture and matrix domains in the form:

$$\text{Fracture: } \phi_f \frac{\partial M_{i,f}}{\partial t} + \nabla \cdot F_i + \Gamma_i = 0 \quad , \quad i = 1, \dots, n_c \quad , \quad (3)$$

$$\text{Matrix: } \phi_m \frac{\partial M_{i,m}}{\partial t} - \Gamma_i = 0 \quad , \quad i = 1, \dots, n_c \quad . \quad (4)$$

The subscripts f and m denote fracture and matrix domains, respectively. The parameter ϕ represents the porosity, M is the overall molar density, t is time, and F_i is the overall molar flux of component i . Γ_i represents the matrix/fracture transfer function, for component i , where the mass transfer includes the combined contributions from advection and diffusion:

$$\Gamma_i = \sum_j \Gamma_{ij} = \sum_j (\Gamma_{ij,a} + \Gamma_{ij,d}) \quad , \quad i = 1, \dots, n_c \quad . \quad (5)$$

Here the subscripts a and d denote the mass transfer due to advection and molecular diffusion, respectively, and subscript j refers to the phases. The evolution of a pressure gradient provides the driving force for fluid advection while concentration gradients provide the driving force for molecular diffusion.

As will be shown in the next sections, the implementation of the DPM into the reservoir simulator is facilitated by the simple (and similar) representations of the mass transfer of both processes (following the functional form of Warren and Root (1963)): Both mechanisms are essentially described by the similar form of the partial differential equation:

$$\frac{\partial X}{\partial t} = \nabla \cdot (D \nabla X) \quad , \quad (6)$$

where for a pressure equation, X (scalar) represents pressure and D (scalar) is hydraulic diffusivity, while for a molecular diffusion problem, X (vector) represents concentration and D (matrix) defines the molecular diffusion coefficients.

Transfer Function for Fluid Expansion/Advection

To express the transfer function due to fluid expansion/advection, we use the generalized Vermeulen form (abbreviated here as gVer) introduced by [Zhang et al. \(2022\)](#) for single-phase flow and later extended to a multiphase problem by [Raslan et al. \(2023b\)](#). The transfer function for component i in phase j is given by:

$$\Gamma_{i,j,a} = \sigma_a \beta_a \rho_j \lambda_j z_{ij} (P_{f,j} - \bar{P}_{m,j}) \quad , \quad i = 1, \dots, n_c \quad , \quad (7)$$

where the parameter ρ_j is the fluid density of phase j , λ_j is the mobility of phase j , and P_j is the pressure of phase j . In the discussion of the representation of two transport mechanisms, it becomes important to emphasize the definition of the time-dependent correction factor, in this case β_a is given by:

$$\beta_a = \left[\frac{1}{V_a} \frac{P_f - P_0}{P_f - \bar{P}_m} \left(\left(\frac{\bar{P}_m - P_0}{P_f - P_0} \right)^{1-V_a} - \frac{\bar{P}_m - P_0}{P_f - P_0} \right) \right] \quad . \quad (8)$$

In [Eq. \(8\)](#), the subscript 0 denotes the initial state and V_a is the model parameter. Unlike the time-dependent correction term for diffusion problems, as presented in the next section, β_a is a scalar quantity. It is worth noting that [Eq. \(7\)](#) reverts to the form of [Warren and Root \(1963\)](#) when pseudo-steady state is approached, i.e., β_a equals to 1.

Transfer Function for Molecular Diffusion

In DP modeling, the transfer function parameters are critical to properly represent the recovery mechanisms. In this paper, a part of our efforts aims to understand whether σ and β are common between viscous flow and molecular diffusion problems.

Expressions analogous to [Eqs. \(7\) and \(8\)](#) cannot be obtained, directly, for molecular diffusion. That is, a time-dependent correction factor for molecular diffusion cannot be derived by a direct transformation from the pressure-diffusion formulation. In the equations above, the pressure diffusion equation was solved to arrive at a common correction factor for all components in a mixture. On the other hand, molecular diffusion is a vector quantity, and while the diffusion equations can be solved for a binary system, it is more complicated to obtain a general solution for a multicomponent system. Therefore, when a full diffusion coefficient matrix is used ([Wu and Jessen, 2020](#)) and the flux depends on the concentration gradients of other components, as for the generalized Fick's law, a decoupling method is required (see [Sherafati and Jessen, 2018](#)). However, in the context of the classical Fick's law, we use a diagonal matrix with elements that represent the effective diffusion coefficient of a component i in the mixture, and the diffusive flux of a component is driven by the related component composition gradient, thereby simplifying the representation.

We can write the diffusive rate for component i in phase j for a multicomponent mixture as:

$$\Gamma_{i,j,d} = \sigma_d \beta_{i,d} \rho_j \frac{\phi}{\tau} S_j D_{i,d} (x_{i,f} - \bar{x}_{i,m}) \quad , \quad i = 1, \dots, n_c \quad . \quad (9)$$

The parameter τ is the tortuosity and S_j is the fluid saturation in phase j .

In this study, to calculate $D_{i,d}$, we use the extended Sigmund correlation to obtain binary diffusion coefficients in the mixture ([Sigmund, 1976a, 1976b; da Silva and Belery, 1989](#)) and the formulation of [Wilke \(1950\)](#) to obtain the effective diffusivity.

Finally in [Eq. \(9\)](#), $\beta_{(i,d)}$ is an element in a n_c vector that represents the time-dependent correction factor per component in the mixture and can be written as follows:

$$\beta_{i,d} = \left[\frac{1}{V_d} \frac{x_f - x_0}{x_f - \bar{x}_m} \left(\left(\frac{\bar{x}_m - x_0}{x_f - x_0} \right)^{1-V_d} - \frac{\bar{x}_m - x_0}{x_f - x_0} \right) \right] \quad , \quad i = 1, \dots, n_c \quad , \quad (10)$$

In DP modeling, the parameters of the generalized Vermeulen transfer function are critical to properly represent the transport mechanisms. Given the similarity in the mathematical representation of viscous flow

(pressure diffusion) and molecular diffusion, we start by investigating if the matrix/fracture interaction parameters are common between viscous flow and molecular diffusion problems, i.e., if $\sigma_e = \sigma_d$ and $V_e = V_d$ (see Eqs. 7–8 and Eqs. 9–10).

In this paper we consider two options for modeling matrix/fracture mass transfer, i.e., Eq. (5): The first option is to use Eqs. (7)–(10) with $\beta=1$ and shape factors from Lim & Aziz (1995), and the second option is to utilize the generalized Vermeulen formulation in Eqs. (7)–(10). The relevant model parameters, for either option, are tabulated in Appendix A (Table A-1).

Numerical Examples

We present several examples to demonstrate the performance of the new DP compositional model. First, we validate the transfer function for a binary diffusion problem in a simple fracture geometry by comparing our calculation results with an analytical solution. Next, we demonstrate the accuracy of the new transfer function for a ternary fluid system with composition-dependent diffusion coefficients. Then, we provide a method to predict matrix/fracture interaction parameters for more complex, realistic fracture geometries. Finally, we apply the DP compositional model to interpret a HnP experiment on an Eagle Ford core.

Example 1: 1D Diffusion in a Binary System

In this example, we demonstrate the accuracy of the diffusion model in the DP representation by comparing calculation results with those from a fine-grid model and the corresponding analytical solution, with a focus on accurately modeling the early-time transient behavior.

We consider a conceptual 1D diffusion problem of a binary system composed of C_1 and CO_2 (Hoteit, 2013). We consider a 10-ft (3 m) long domain which can be represented by 1) a fine-grid SPM (100 cells of equal sizes) and 2) an equivalent single-block DPM. For the SPM, we assume constant composition boundary conditions of 100% CO_2 and the remainder of the domain saturated with C_1 , creating a composition discontinuity in the binary system (see Fig. 1). A similar setup is defined for the single-block DPM: the ‘fracture’ cell initially contains 100% CO_2 , while the ‘matrix’ cell contains 100% C_1 . The 1D problem is represented by a DPM with one set of parallel fractures and we consider two options for the transfer function: The traditional transfer function using shape factors from Lim & Aziz (1995), and the generalized Vermeulen transfer function.

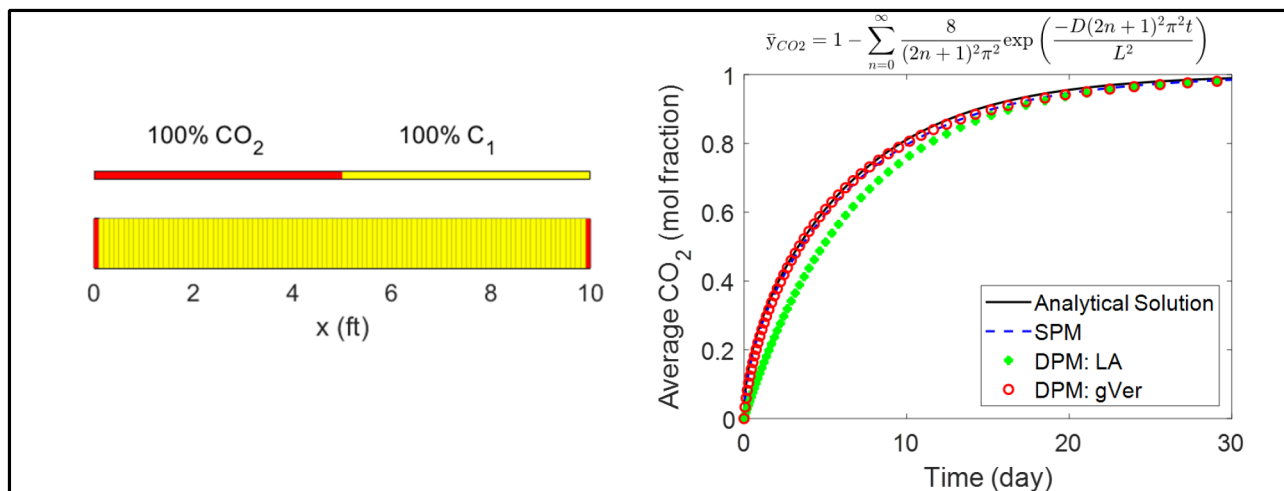


Figure 1—1D domain for a binary system (left panel). Average CO_2 mole fraction for four solutions (analytical solution, fine-grid SPM, and 2 single-block DPM's) in the right panel.

We consider diffusion at standard conditions (60 F/288 K and 14.7 psi/1 atm) where CO_2 and C_1 behave like ideal gases. For all the solutions, we assume a constant diffusion coefficient of $1.5\text{e-}5 \text{ ft}^2/\text{s}$. Finally, the corresponding analytical solution (Crank, 1975) is also reported in Fig. 1 – right panel.

Figure 1 (right panel) reports the average composition of CO_2 for all four solutions: the analytical solution, the fine-grid SPM, the DPM with traditional transfer function (DPM: LA), and the DPM with the gVer (DPM: gVer) transfer function. Based on the observations from this example, two key remarks can be made:

1. All numerical solutions agree with the analytical solution except for the traditional transfer function, which only agrees with the other solutions after about 14 days (when the average concentration reaches ~ 0.9).
2. The model parameters used in the generalized Vermeulen transfer function (σ and V) are similar to that for an equivalent viscous flow problem.

Example 2: Ternary System with Variable Diffusion Coefficients

Next, we test the new formulation for a ternary problem with composition-dependent diffusion coefficients. In the context of the DPM, this example emphasizes the importance of using a unique time-dependent correction factor for each component in the mixture.

We consider the domain from Example 1 and consider a ternary fluid (CO_2 , C_1 , and C_2) with initial molar compositions reported in Fig. 2 (at standard conditions) and a constant boundary composition of 100% CO_2 . The diffusion coefficients vary according to their compositions as predicted by the correlation of Sigmund (1976) and Wilke (1950).

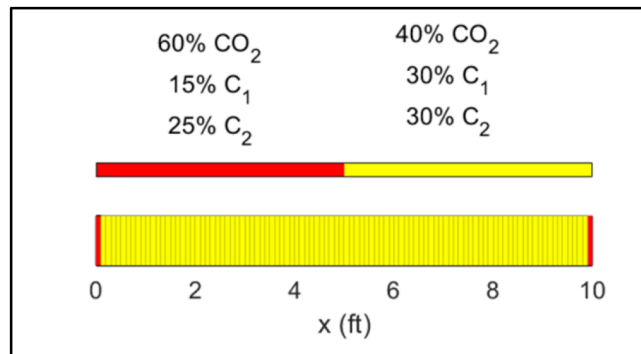


Figure 2—1D domain (similar to that in Example 1) showing the initial molar composition.

The average CO_2 and C_1 compositions are reported in Fig. 3. As can be seen, the results are consistent with those for the binary example: The generalized Vermeulen transfer function agrees with the reference SPM solution when using the same matrix/fracture transfer function parameters as for a binary system.

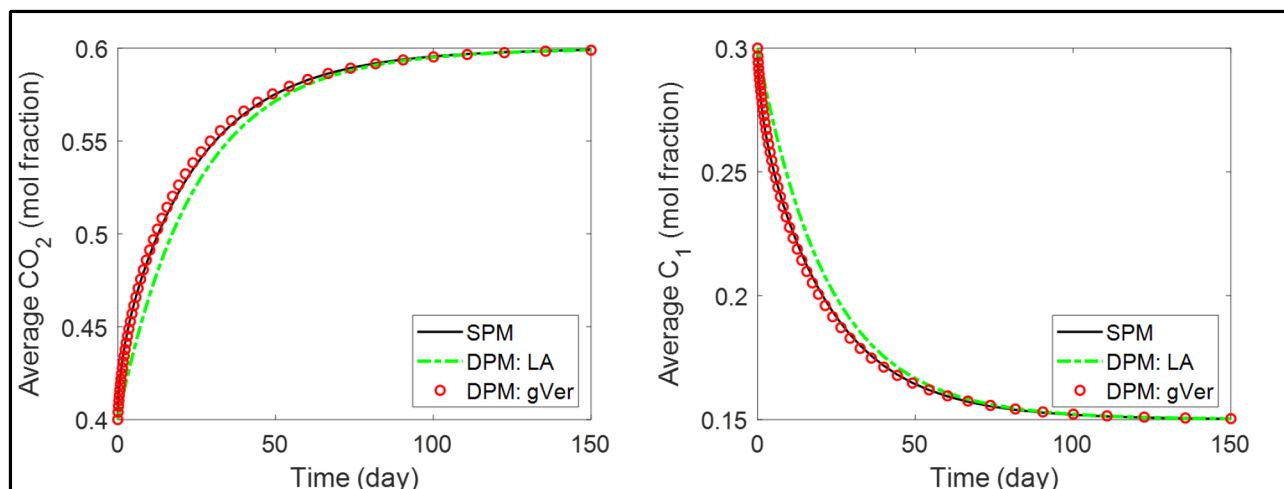


Figure 3—Average CO_2 (left panel) and C_1 (right panel) compositions for the three solutions: fine-grid SPM, DPM + Lim & Aziz, and DPM + gVer

We also report the average effective diffusivity per component, and the time-dependent correction factor in Fig. 4. In multicomponent diffusive mass transfer, as considered in this example, the transfer rate per component reflects the physics of the problem: The rate of transfer for each component is a function of its diffusivity and concentration gradient.

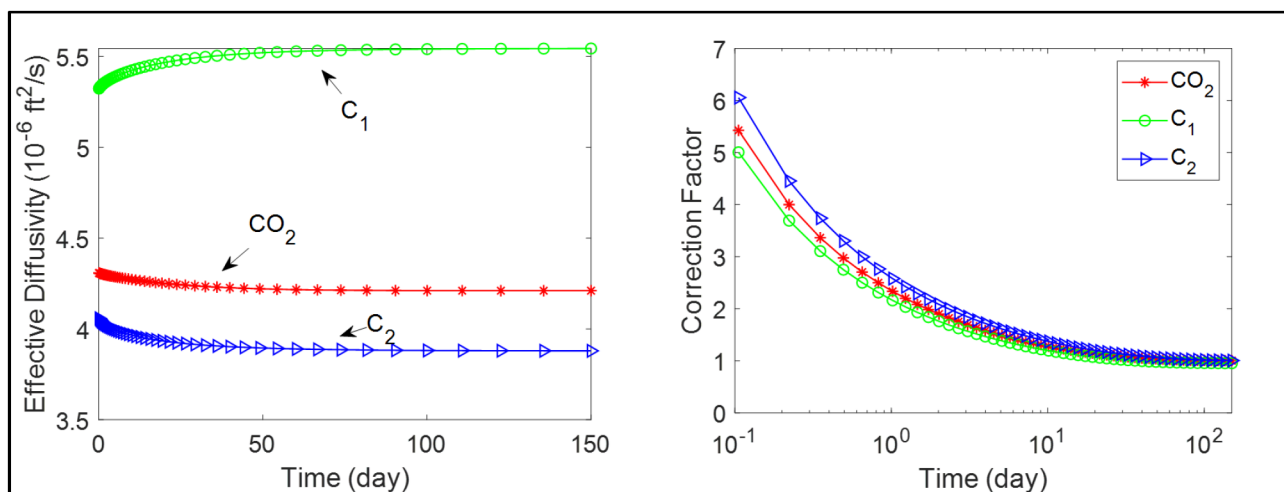


Figure 4—Average effective diffusion coefficients for the three components over time (left panel). Time-dependent correction factor, per component, used in the generalized Vermeulen transfer function (right panel)

Example 3: Two Sets of Parallel Fractures (Asymmetric)

The general perception is that the DPM provides a sugar-cube representation of fractured systems where the matrix/fracture geometries are idealized and associated with analytical solutions. However, [Raslan et al. \(2023a\)](#) demonstrated that DPM's can capture the fluid flow behavior for complex fracture problems. In this example, we continue to demonstrate the advantage of the DPM for irregularly shaped matrix/fracture geometries in calculations of diffusive mass transfer.

We consider the same domain used in [Raslan et al. \(2023a\)](#) with two sets of parallel (orthogonal) fractures (see Fig. 5). We assume the matrix (30% porosity and 5 nD permeability) to contain 100% C_1 and that the fractures contain a constant composition of 100% CO_2 . To solve this problem, we need to define the relevant transfer function parameters. It was documented in [Raslan et al. \(2023a\)](#) that the standard (or pre-determined) transfer function parameters (reported in Table A-1) are insufficient to describe the

geometry of this system. Therefore, to obtain representative model parameters, we first model the setup as a single-component pressure depletion problem and fit the transfer function parameters to the SPM reference solution.

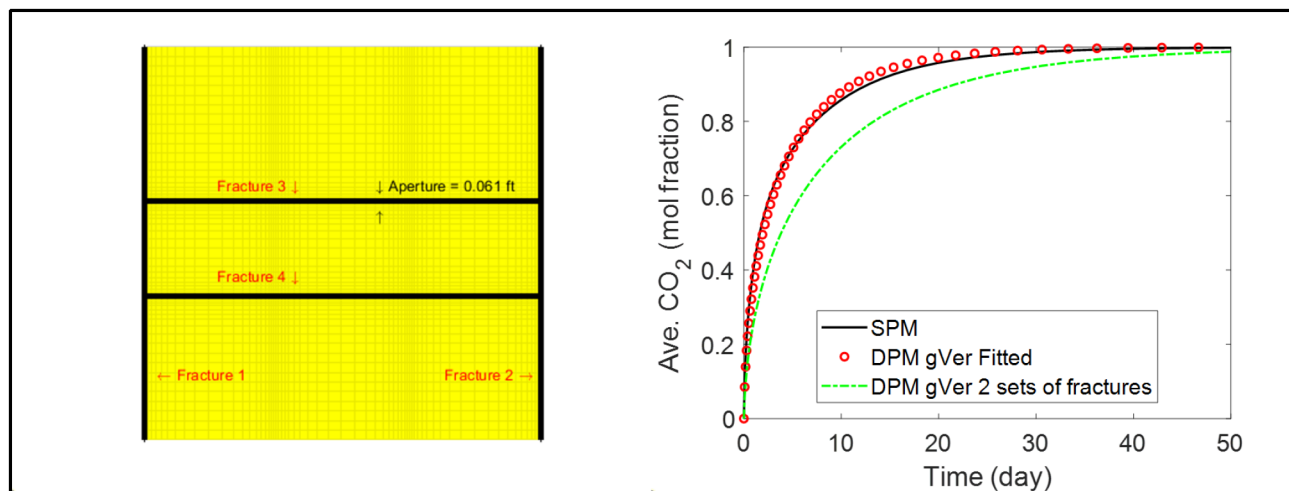


Figure 5—Top view of the domain that includes two sets of parallel (orthogonal) fractures (left panel). Average CO₂ mole fraction for three solutions: fine-grid SPM, DPM + gVer fitted, and DPM + gVer assuming 2 sets of fractures.

As presented in [Raslan et al. \(2023a\)](#), the domain was modeled with DPM and fitted to the fine-grid SPM solution. Using the resulting values of $\sigma_e = 0.11 \text{ ft}^2$ and $V_e = 2.04$, we conduct the calculation of the diffusion problem and report the results in [Fig. 5](#) – right panel. It is observed that a calculation with DPM + gVer, using pre-determined parameters, fails in reproducing the reference solution, whereas the fitted DPM + gVer matches the reference SPM solution accurately. This suggests that for a diffusion problem, an equivalent pressure problem can be used to obtain the relevant matrix/fracture transfer function parameters, which becomes useful (preferred) for multicomponent diffusion problems.

Example 4: Modeling of CO₂ HnP Experiment on Eagle Ford Shale Core

The role of molecular diffusion becomes relevant in problems of EOR: During HnP processes in tight fractured systems, CO₂ diffuses into the oil allowing the oil to swell and mobilize. This becomes particularly prominent during the soaking period of HnP operations.

In this example, we present our efforts to interpret CO₂ HnP in an Eagle Ford core via MRST compositional calculations (the details of the core as well as the lab experiment procedures fall outside the scope of this manuscript and are presented in a separate manuscript). Here, we focus exclusively on the modeling of the relevant experiments. Two approaches are used to represent the core: a fine-grid SPM and a single-block DPM. In both representations, the classical Fick's law ([Sigmund, 1976](#); [Wilke, 1950](#)) was used to predict the diffusivities. The objective is to demonstrate the accuracy and simplicity of the DPM (with gVer), relative to the reference SPM, in complex problems involving diffusive and viscous flow recovery mechanisms. As such, in this example we emphasize the soaking period of the HnP process wherein the exchange of fluids due to diffusion is significant.

Construction of the Numerical Domain and the Fluid System. A compositional SPM was built in MRST using a radial grid to represent an Eagle Ford core (see [Fig. 6](#)). The extraction cell, mimicking the behavior of a larger fracture, is represented by the yellow outer gridblocks, and the core is shown with red gridblocks. Grid refinement, in radial and vertical directions, was included to reduce numerical artifacts. The porosity of the extraction cell was set to 99% to mimic the void space, and a ‘large’ value of permeability equal to 0.500 Darcy was assigned. The matrix porosity (11.9%) and matrix permeability $k_x = k_y = 2.79 \text{ μD}$ and $k_z = 3.88 \text{ μD}$

were assumed constant throughout the domain. A tortuosity factor of 6 was used here to represent diffusion in the porous material.

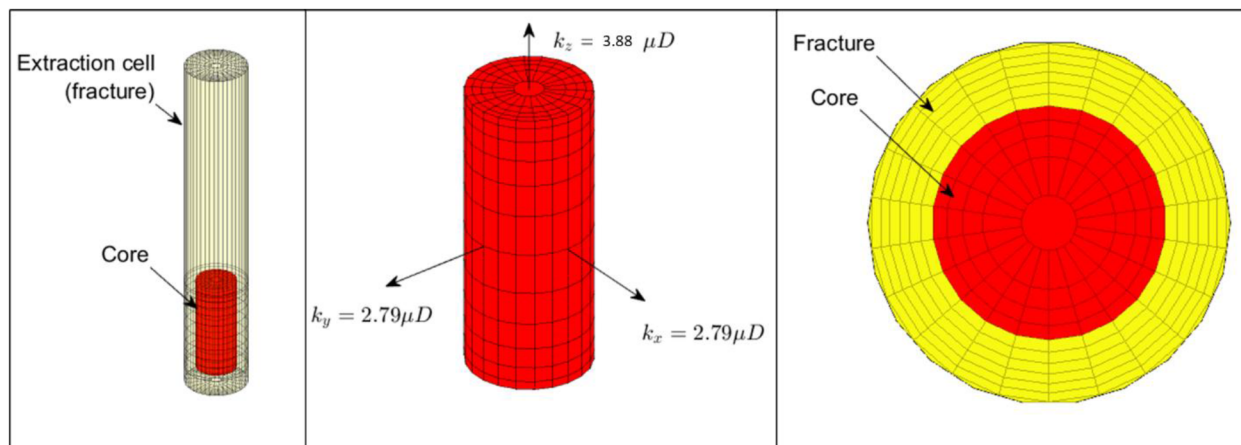


Figure 6—Numerical representation of the core using MRST. 3D view of the extraction cell and the core (left panel), 3D view of the core (middle panel), and horizontal cross-section at the top of the core (right panel).

We also construct a single-block DPM that mimics the SPM (see Fig. 7). Although that may appear to be an oversimplified representation, it allows us to easily and distinctly assign the physical properties of the extraction cell and the core. A tortuosity value of 6 was also used for the DPM, while the transfer function parameters are discussed in more detail below.

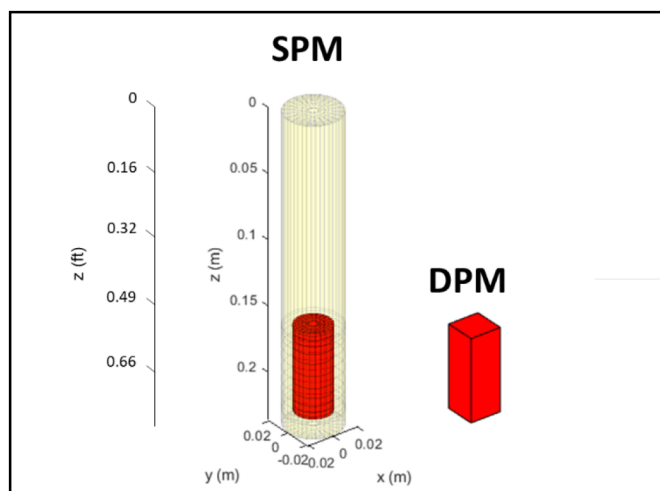


Figure 7—Numerical representations of the EF-3 core in SPM and DPM.

The equation-of-state (EOS) modeling of the relevant fluid system, including mixtures of the synthetic oil with CO₂, is discussed in Alahmari et al. (2023), and we summarize here the EOS input in Table 1.

Table 1—EOS parameters and interaction coefficients for synthetic oil and CO₂

Component	P_c (psia/ kPa)	T_c (F/K)	ω	$shift$	$k_{CO_2,j}$	z_i , mol%
CO ₂	1070/7,376	87.9/304.2	0.23	0.12279	0	-
nC ₁₀	306/2,108	652.0/617.6	0.49	0.06478	0.09694	7.90
nC ₁₂	265/1,824	725.1/658.2	0.56	0.10308	0.08947	19.9
nC ₁₄	235/1,621	789.5/694.0	0.68	0.15413	0.08777	30.8
nC ₁₆	206/1,419	830.9/717.0	0.74	0.17669	0.07676	41.4

The core model was initialized with 100% synthetic oil while the extraction cell was initialized with 100% CO₂ at 14.7 psia/101.32 kPa and a temperature of 158 F/343.15 K. Fluid injection and extraction are represented by two wells completed at the top and bottom of the (extraction cell) grid, respectively. Quadratic gas/liquid relative permeability functions were assumed for the matrix while straight-line relative permeability functions were used for the extraction cell (fracture).

Selection of Matrix/Fracture Model Parameters. The purpose of this study is to provide guidance on the selection of the transfer function parameters. We therefore proceed with an overview of the process to provide clarity on the selection. We consider the numerical representations described in Fig. 7 and focus initially on a single-component (C₁) pressure depletion problem (temperature of 60 F/288 K). We start at 7,000 psia/48 MPa and deplete the system at 2,500 psia/17.2 MPa. The results for different modeling approaches are reported in Fig. 8 below:

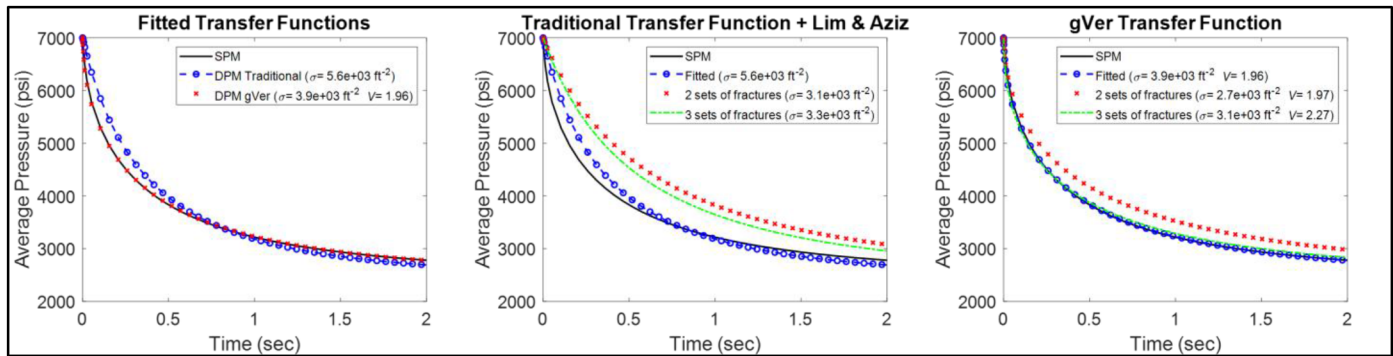


Figure 8—Average pressure during depletion: Left - fine-grid SPM and fitted DPM's (traditional and advanced transfer function). Middle panel - SPM and the traditional transfer function. Right - SPM and the gVer transfer function. For all cases, the relevant matrix/fracture parameters are reported in the legend.

As observed in the left panel of Fig. 8, the fitting (shape factor) of the traditional transfer function to the reference SPM solution doesn't provide for an accurate response. However, the advanced transfer function matches with the reference solution accurately with values of $\sigma_e = 3.9e3 \text{ ft}^{-2}/4.2e4 \text{ m}^{-2}$ and $V_e = 1.96$. Furthermore, the middle and right panels report the pressure response assuming 2 and 3 sets of parallel fractures, for both transfer functions. As can be seen the traditional transfer function with Lim & Aziz (1995) shape factors provide limited accuracy for the average pressure. While the response of the gVer with pre-determined σ_e and V_e ($3.1e3 \text{ ft}^{-2}/3.3e4 \text{ m}^{-2}$ and 2.27, respectively) assuming 3 sets of orthogonal fractures overlaps with the fitted parameters (more on this in the next section).

We note that the use of the Cartesian grid in the DPM representation (Fig. 7 above) is to allow for a simpler/direct application of the transfer function parameters from Zhang et al., (2022), though it is feasible

to generate new values for σ and V using analytical solutions for relevant cylindrical geometry from, i.e., 1D radial flow - Crank (1975).

DPM Representation of the Soaking Period. Next, we consider the application of the generalized Vermeulen transfer function to the HnP problem. It is assumed that the CO₂ injection, the first step of the HnP process, is completed and we focus on the soaking period. After the completion of the injection period, the system (extraction cell + core) is now at 4,160 psia/28.68 MPa with the fluid composition (mol%) shown in Table 2.

Table 2—Composition in the system (extraction cell + core) at the start of the soaking period.

CO ₂	nC ₁₀	nC ₁₂	nC ₁₄	nC ₁₆
99.7	0.026	0.068	0.11	0.14

A sharp gradient at the matrix/fracture interface is present where the fracture contains 100% CO₂ while the core contains a mixture of oil and CO₂ (see Fig. 9). This allows for the initiation of diffusive mass transfer between the two domains driven by the concentration gradients and the effective diffusivities.

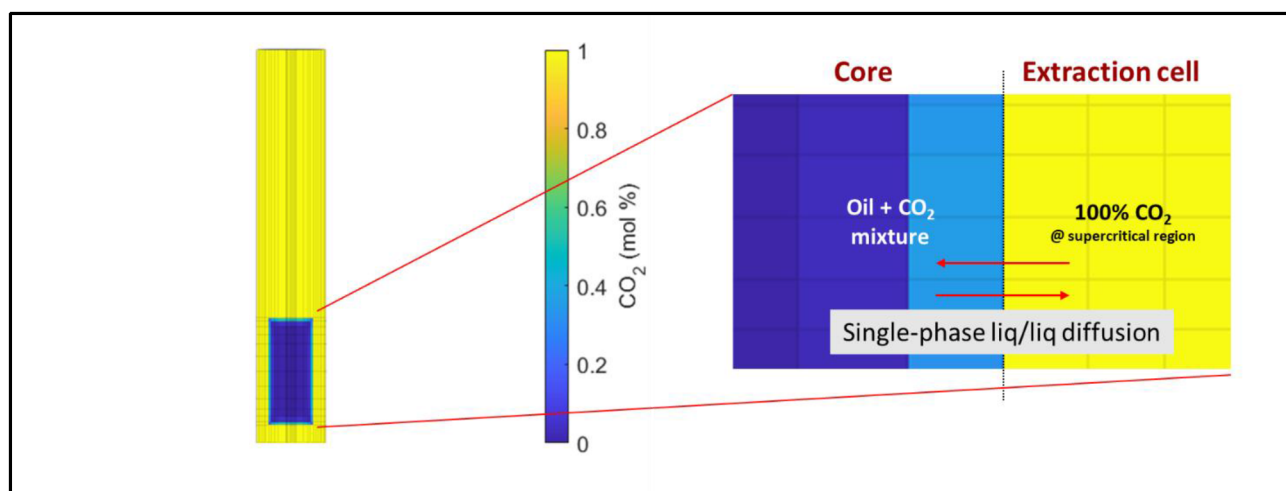


Figure 9—Matrix/fracture interface compositions. Left: vertical cross-section of the model, and Right: a close-up of the matrix/fracture boundary, both panels shown at the start of soaking.

We apply different matrix/fracture transfer function parameters to demonstrate the importance of proper selection to model the diffusive transport between the two domains. Figure 10 reports the average pressure response from the fine-grid SPM (red crosses) and 4 DPM calculations with the generalized Vermeulen transfer function. All DPM calculations assume $\sigma_d = \sigma_e = \sigma$ and $V_d = V_e = V$, however, with different approaches for obtaining these values: 1) DPM + gVer fitted to the single-component pressure depletion problem (maroon circles); 2) DPM + gVer assuming 2 sets of fractures (yellow triangles); 3) DPM + gVer assuming 3 sets of fractures (purple stars); and 4) DPM + gVer fitted to the SPM HnP calculation (dashed green).

As observed in Fig. 10, using the single-component pressure depletion problem for evaluating the transfer function parameters, does not provide for an accurate result. Furthermore, the DPM calculations assuming 2 and 3 sets of orthogonal fractures show similar departure from the reference SPM solution. The last set of σ and V were obtained by matching the pressure response of the DPM to the fine-grid SPM during the soaking period; these new values are $\sigma = 8.61e3 \text{ ft}^{(-2)}/9.3e4 \text{ m}^{(-2)}$ and $V = 1.96$.

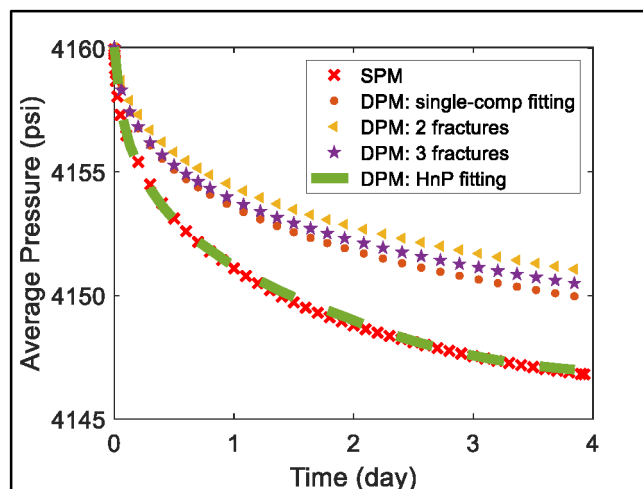


Figure 10—Average pressure response during soaking period for the fine-grid SPM and different DPM's.

It is evident from this example that *a priori* knowledge of the recovery mechanism(s) is needed for setting up the proper transfer function. The diffusion flux versus advective flux from the above calculations are shown in Fig. 11 below. As can be seen, the first 3 examples predict a diffusion-dominated transport; i.e., advective transport is at least 3 orders of magnitude less than the diffusive transport. However, in Example 4, the advective transport is more significant (though still less than the diffusive transport), as expected during the soaking phase of a HnP process: In this case, the values of σ and V determined from a simplified depletion calculation, as shown previously, does not provide for a good agreement with the HnP process.

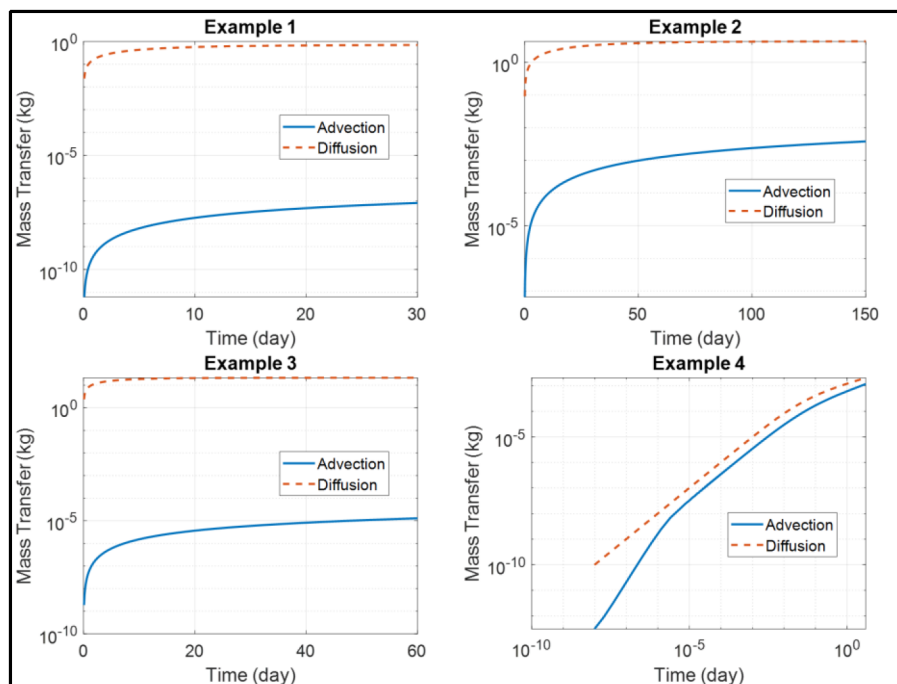


Figure 11—Mass transfer (diffusive and advective) for all examples.

Another point of emphasis from Fig. 10 is the difference in response between the two methods: DPM with parameters obtained from the single-component depletion and those obtained assuming 3 sets of fractures. The difference between these two solutions is not what was observed in the previous examples (seen in Fig.

8 – right panel). In the context of EOR, this can be attributed to the directionality of the diffusion process. Additional work is underway to further illustrate this point.

Conclusion and Summary

This paper proposes a new DPM to investigate diffusive mass transfer in tight fractured reservoirs, selected for its modest computational requirements and accurate representation of the physics at play. A key innovation is the introduction of an enhanced matrix/fracture representation that incorporates both advective and diffusive mass transfer mechanisms for a wide range of geometries and complex EOR operations. This approach leverages the Warren and Root (1963) type model, augmented by a time-dependent correction factor derived from Zhang et al. (2022), allowing for more accurate calculations in tight systems.

Numerical validations and benchmarks, including comparisons with analytical solutions and fine-grid SPM simulations, confirm the robustness of the proposed DP compositional model in diffusion-dominated environments for both binary and multicomponent fluid systems. For all presented examples, the transfer function parameters are the same for diffusive and advective problems. Depending on the problem, where advective flux is negligible, the model parameters for the diffusion problem can be obtained from a simpler/equivalent pressure depletion problem. However, when advective flux is nonnegligible, such as in the case of the HnP example, the matrix/fracture interaction parameters need to be obtained from the soaking period to represent both mechanisms at play.

Although the DPM may seem an oversimplified representation, it allows for simple modeling of fractured system, enabling the study of complex EOR processes. Its ability to overcome the limitations of conventional modeling approaches and reduce the need for fine gridding provides for a practical solution in larger-scale EOR applications, paving the way for broader industrial adoption and future research.

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Appendix-A

Transfer Function Input Parameters

Table A-1 summarizes the transfer function parameters for 1, 2, and 3 sets of fractures for use in dual-porosity modeling: 1) Shape factors from Lim and Aziz (1995) to be used in with a time-dependent correction factor of 1; 2) Parameters for the generalized Vermeulen transfer function (Zhang et al., 2022), σ and V .

Table A-1—Tabulated below are Lim and Aziz (1995) and Generalized Vermeulen (2022) transfer function parameters.

Set(s) of fractures	Lim And Aziz (1995)	Generalized Vermeulen (2022)	
	σL^2	σL^2	V
1	π^2	8.76	1.54
2	$2\pi^2$	17.44	1.97
3	$3\pi^2$	27.49	2.27